FPPF HYPERCOVERINGS: A^* IS AN EQUIVALENCE ON QUASI-COHERENT MODULES WITH QUASI-INVERSE A_*

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the fully-faithful transfer is obtained by what the author calls 'a formal argument (chasing around the diagram)' (source line 7594) - the same glue step documented on cards L46151, L46161 and L46163, with all inputs (full faithfulness of h^* and a_{fppf}^* with quasi-inverses, the commutative ringed diagram of card L46165) established before the sentence; the closing computation F = h_{-1,*}a_{fppf,*}h^*G = a_*h_*h^*G = a_*G uses the componentwise descriptions and h_*h^* = id on the essential image; no verification is omitted.

Editorial note. The label lemma-descent-qcoh-for-fppf-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The ringed comparison diagram of the fppf-etale sites. Same chapter, \label{lemma-compare-simplicial-objects-fppf-etale-modules} (card L46165 of this batch). Lines 7557-7560: the diagram h with h_{-1} = a_X and components h_n, as morphisms of ringed topoi.
- QCoh equivalence along the big-small comparison. More on Cohomology of Spaces chapter, \label{spaces-more-cohomology-lemma-review-quasi-coherent}. Lines 7561-7567: (h_{-1})^* : QCoh(O_X) -> QCoh(O_big) is an equivalence with quasi-inverse h_{-1,*}, likewise componentwise.
- Quasi-coherent modules on the total site characterized. Same chapter, \label{lemma-quasi-coherent-sheaf} (card L46152 of this batch). Lines 7568-7571 and 7590-7593: QCoh(O_U) and QCoh(O_{big,total}) are the cartesian modules with quasi-coherent components.
- The simplicial-site module functors agree on components. Same chapter, \label{lemma-morphism-simplicial-sites-modules}. Lines 7571-7578: h^* and h_* act componentwise, lifting the component equivalences to h^* : QCoh(O_U) -> QCoh(O_{big,total}).

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Simple descent for modules along hypercoverings on the big fppf site. Same chapter, \label{lemma-hypercovering-X-simple-descent-modules}. Lines 7581-7585: a_{fppf}^* fully faithful with quasi-inverse a_{fppf,*} and essential image the cartesian modules.

Source reference. Original reference 0DHC: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7487–7487. Prerequisite pointer invoked at proof line(s) 7559; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DAB: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5209–5209. Prerequisite pointer invoked at proof line(s) 7592; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1604–1604. Prerequisite pointer invoked at proof line(s) 7580; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7M: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2817–2817. Prerequisite pointer invoked at proof line(s) 7578; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 09A0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5012–5012. Prerequisite pointer invoked at proof line(s) 7591; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7554; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DGN: sources/stacks-chow/tags.txt, source lines 16878–16878. Prerequisite pointer invoked at proof line(s) 7566; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DGM: sources/stacks-chow/tags.txt, source lines 16877–16877. Prerequisite pointer invoked at proof line(s) 7563; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. If $a : U \to X$ is an fppf hypercovering of X, then*

$$a^* : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_U)$$

is an equivalence fully faithful with quasi-inverse given by a_*. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.

Proof. Consider the diagram of Lemma 0DHC. In the proof of this lemma we have seen that h_{-1} is the morphism a_X of More on Cohomology of Spaces, Section 0DGM. Thus it follows from More on Cohomology of Spaces, Lemma 0DGN that

$$(h_{-1})^* : QCoh(\mathcal{O}_X) \longrightarrow QCoh(\mathcal{O}_{big})$$

is an equivalence with quasi-inverse $h_{-1,*}$. The same holds true for the components h_n of h. Recall that $QCoh(\mathcal{O}_U)$ and $QCoh(\mathcal{O}_{big,total})$ consist of cartesian modules whose components are quasi-coherent, see Lemma 0D7M. Since the functors h^* and h_* of Lemma 0DH0 agree with the functors h_n^* and $h_{n,*}$ on components we conclude that

$$h^* : QCoh(\mathcal{O}_U) \longrightarrow QCoh(\mathcal{O}_{big,total})$$

is an equivalence with quasi-inverse h_*. Observe that U is a hypercovering of X in $(Spaces/S)_{fppf}$ as defined in Section 09A0. By Lemma 0DAB we see that a_{fppf}^* is fully faithful with quasi-inverse $a_{fppf,*}$ and with essential image the cartesian sheaves of $\mathcal{O}_{fppf,total}$-modules. Thus, by the description of $QCoh(\mathcal{O}_{big})$ and $QCoh(\mathcal{O}_{big,total})$ of Lemma 0D7M, we get an equivalence

$$a_{fppf}^* : QCoh(\mathcal{O}_{big}) \longrightarrow QCoh(\mathcal{O}_{big,total})$$

with quasi-inverse given by $a_{fppf,*}$. A formal argument (chasing around the diagram) now shows that a^* is fully faithful on $QCoh(\mathcal{O}_X)$ and has image contained in $QCoh(\mathcal{O}_U)$.

Finally, suppose that $\mathcal{G}$ is in $QCoh(\mathcal{O}_U)$. Then $h^*\mathcal{G}$ is in $QCoh(\mathcal{O}_{big,total})$. Hence $h^*\mathcal{G} = a_{fppf}^*\mathcal{H}$ with $\mathcal{H} = a_{fppf,*}h^*\mathcal{G}$ in $QCoh(\mathcal{O}_{big})$ (see above). In turn we see that $\mathcal{H} = (h_{-1})^*\mathcal{F}$ with $\mathcal{F} = h_{-1,*}\mathcal{H}$ in $QCoh(\mathcal{O}_X)$. Going around the diagram we deduce that $h^*\mathcal{G} \cong h^*a^*\mathcal{F}$. By fully faithfulness of h^* we conclude that $a^*\mathcal{F} \cong \mathcal{G}$. Since $\mathcal{F} = h_{-1,*}a_{fppf,*}h^*\mathcal{G} = a_*h_*h^*\mathcal{G} = a_*\mathcal{G}$ we also obtain the statement that the quasi-inverse is given by a_*. □